Anhnamxtanh at HalluScoring-2026: Counterfactual Cross-Checking for Hallucination Detection and Answer Verification in Islamic Knowledge

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Abstract

Hallucination detection in Arabic Islamic question answering is challenging because fluent answers may misstate a verse, narration, ruling, attribution, or historical fact. The task jointly requires detecting hallucination and recovering the factually correct option. We introduce C3R, a Counterfactual Cross-Check Recognition framework that couples these outputs through two role-conditioned predictions from one Arabic-centric language model: a truth role selects the factual option, while a match role selects the candidate that best matches the generated response. Training uses targeted counterfactual responses, hard distractors, option-order permutations, and completion-only supervision; no cross-view consistency loss is used. At inference, C3R scores valid structured completions, maps option probabilities from permuted presentations back to semantic identities, averages them, and derives the hallucination label from argmax agreement between the two roles. The cross-check is therefore applied only at inference. With a QLoRA-adapted ALLaM-7B backbone, C3R ranks first on the official test leaderboard with a final score of 0.8824, comprising 0.8673 macro-F1 and 0.8975 option accuracy. These results show that structured cross-role agreement can support strong performance in knowledge-sensitive hallucination detection and answer verification.

1 Introduction

Large language models (LLMs) often produce fluent but unsupported content, commonly described as hallucination (Ji et al., 2023; Huang et al., 2025). This is especially consequential in Arabic Islamic question answering, where a minor error in a name, verse, attribution, ruling, or date can create misinformation. Arabic resources document this problem (Mubarak et al., 2024), yet surface fluency remains weak evidence of factuality. A response may remain topical and grammatical while changing a verse number, legal condition, or chain of attribution. The setting therefore requires fine-grained distinctions among closely related religious and historical statements.

We study the Islamic-knowledge track of the HalluScoring 2026 shared task (Alansari et al., 2026), which requires both a hallucination label and the factually correct answer among six options. These outputs are structurally linked through the candidate that best represents the generated response: agreement with the factual option supports a non-hallucination decision, whereas disagreement supports hallucination. Treating them as independent classification problems discards this consistency constraint. Free-form generation requires mapping unconstrained text back to the fixed label set, while multiple-choice predictions can change under option reordering (Pezeshkpour and Hruschka, 2024; Wei et al., 2024).

To address these issues, we propose C3R, a Counterfactual Cross-Check Recognition framework. A shared model learns a truth role that selects the factual option while treating the generated response as untrusted, and a match role that selects the option expressing that response. Training combines gold-response and hard-negative counterfactuals with option permutations. Inference scores only valid completions, averages probabilities after mapping permuted labels back to their semantic identities, and declares the response non-hallucinated when the two roles select the same option. Here, cross-check denotes this inference-time agreement, not a cyclic training loss.

2 Related Work

Hallucination detection and factuality. Hallucination research covers detection and factual verification for open-ended outputs (Ji et al., 2023; Huang et al., 2025). HaluEval frames response-level discrimination (Li et al., 2023),

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 1154–1159
October 28-29, 2026 ©2026 Association for Computational Linguistics

while FActScore decomposes outputs into atomic claims for evidence-based verification (Min et al., 2023). Evidence-free approaches instead use sampled disagreement, semantic perturbations, cross-model checks, or generated verification questions (Manakul et al., 2023; Zhang et al., 2023; Dhuliawala et al., 2024). These methods generally operate over free-form outputs and may require repeated generation or auxiliary verification. C3R obtains two role-conditioned distributions from one model over a fixed candidate set and derives the label directly from their agreement.

Arabic factuality and language models. Arabic hallucination detection remains comparatively underexplored. Halwasa provides 10,000 Arabic LLM-generated sentences annotated for factual and linguistic errors (Mubarak et al., 2024), and later work evaluates prompted recognition of Arabic hallucinations (Sibaee et al., 2024). HalluTruthQA extends response-level detection to error localization, explanation, and factual verification (Bouchekif et al., 2026), while HalluTruthQA-4K emphasizes expert construction and detailed error annotation (Bekhouche et al., 2026). C3R focuses on the shared-task Islamic setting and couples response-level detection with candidate selection, rather than span localization or explanation generation.

Robustness. Robust prediction can aggregate alternative views, as in self-consistency over reasoning paths (Wang et al., 2022). Multiple-choice models are also sensitive to option order, labels, and token choices (Pezeshkpour and Hruschka, 2024; Wei et al., 2024; Zhou et al., 2024). C3R treats option order as a nuisance variable: training remaps supervision under shuffling, while inference averages aligned option probabilities across deterministic presentations.

3 Counterfactual Cross-Check Recognition

3.1 Problem Formulation and Recognition Roles

Figure 1 summarizes the C3R training and inference pipelines. Each instance is $x = (q, g, \mathcal{O})$, where q is an Arabic question, g is a generated answer, and $\mathcal{O} = \{(k_j, o_j)\}_{j=1}^{K}$ is the candidate set. Here $K = 6$ and $k_j \in \{\mathrm{A}, \ldots, \mathrm{F}\}$. Let $a^\star$ be the factually correct option and let $m^\star = m(g, \mathcal{O})$ be the option that best matches g. C3R uses one model with two prompts: the TRUTH role treats g as untrusted and targets $a^\star$, whereas the MATCH role targets $m^\star$ without judging whether g is correct. For original examples, $m^\star$ is assigned by combining Arabic-normalized character similarity and token Jaccard overlap with weights 0.6 and 0.4; the counterfactual examples below have exact match targets.

Both roles target the constrained JSON completion $c(k) = \texttt{\{"selected_option": "}k\texttt{"\}}$ and differ only in their instruction and target option. Prompt tokens are masked, and standard autoregressive cross-entropy is applied only to completion tokens in the augmented training set. No cross-view consistency loss is used.

3.2 Counterfactual and Permutation Augmentation

For each labeled instance, C3R creates four variants: (i) the original TRUTH example targeting $a^\star$; (ii) the original MATCH example targeting $m^\star$; (iii) a positive match example with $g \leftarrow o_{a^\star}$ and target $a^\star$; and (iv) a negative match example with $g \leftarrow o_h$ and target h. The hard negative h is the incorrect option most similar to the gold option under the same Arabic-normalized similarity measure. These interventions change the response while keeping the question, candidates, and factual answer fixed.

Each variant is rendered in the original order and one seeded permutation ($R_{\text{train}} = 2$). Option symbols and targets are remapped together, preserving semantic identity and yielding eight supervised sequences per instance.

3.3 Constrained Likelihood Recognition

At inference, C3R scores rather than generates. For each role r, semantic option k, and presentation permutation π, it computes the length-normalized likelihood of the corresponding valid completion:

$$s_{r,k}^{(\pi)} = \frac{1}{|c(\pi(k))|} \sum_{t=1}^{|c(\pi(k))|} \log p_\theta(c_t(\pi(k)) \mid P_r(x^\pi), c_{<t}(\pi(k))) \,. \tag{1}$$

The scores are converted to option probabilities using a temperature-scaled softmax. After mapping display symbols back to semantic option identities, C3R averages the distributions over R_{test} deterministic presentations:

$$\bar{p}_{r,k} = \frac{1}{R_{\text{test}}} \sum_{\pi \in \Pi} p_{r,k}^{(\pi)}. \tag{2}$$

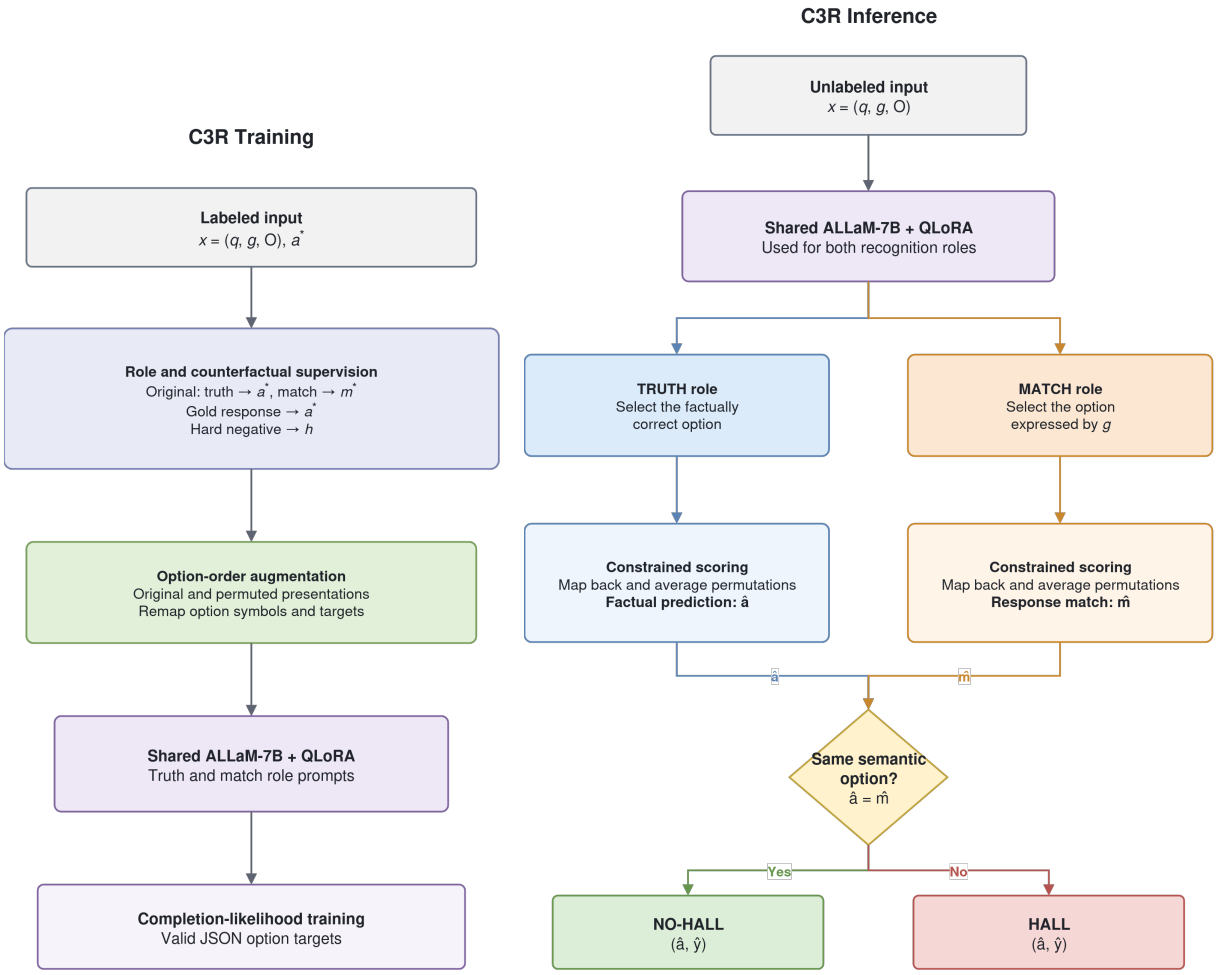


Figure 1: Separate C3R pipelines for counterfactual training (left) and cross-role inference (right).

Enumerating all candidates guarantees an in-set prediction and avoids parsing free-form output.

3.4 Cross-Role Agreement Decision

The two roles produce the factual prediction $\hat{a} = \arg\max_k \bar{p}_{\text{truth},k}$ and response-match prediction $\hat{m} = \arg\max_k \bar{p}_{\text{match},k}$. C3R declares the response non-hallucinated exactly when they identify the same semantic option:

$$\hat{y} = \begin{cases} \text{NO-HALL}, & \hat{a} = \hat{m}, \\ \text{HALL}, & \text{otherwise.} \end{cases} \tag{3}$$

Thus, cross-role agreement is used only as the inference decision rule in Eq. 3.

4 Experiments

4.1 Dataset and Task

We follow the Islamic-knowledge data release and task definition provided by the HalluScoring 2026 shared task (Alansari et al., 2026). The release contains 400 labeled training instances, 200 labeled development instances, and 400 test instances. Each instance consists of an Arabic question, an LLM-generated answer, and six closely related candidate answers indexed from A to F. Training and development instances additionally provide a binary hallucination label and the factually correct option; the corresponding test targets are hidden.

We use the 400 official training instances for adaptation and the 200-instance development set for the analyses reported below. In accordance with the shared-task rules, we use only the official data and no external resources. With one hard negative and two option presentations, the four role-specific variants in Section 3.2 produce 3,200 supervised completion sequences. The official test instances are kept separate and evaluated only through external evaluation.

4.2 Evaluation Metrics

The official score averages two output metrics: macro-F1 for hallucination labels and exact accu-

racy for factual-option predictions over A–F:

$$\text{FinalScore} = 0.5\,\text{MacroF1} + 0.5\,\text{OptionAcc}. \quad (4)$$

Macro-F1 balances the `no_hallucination` and `hallucination` classes, while option accuracy requires an exact match to the gold option. The gold option represents the generated answer for non-hallucinated instances and the corrected factual answer otherwise.

4.3 Implementation Details

C3R uses ALLaM-7B as its backbone (Bari et al., 2025). We adapt the model with QLoRA (Hu et al., 2022; Dettmers et al., 2023) for three epochs using learning rate 10^{-4}, per-device batch size 1, gradient accumulation over 16 steps, and maximum sequence length 2,048. The frozen backbone uses 4-bit NF4 double quantization. LoRA adapters with rank 32, scaling 64, and dropout 0.05 are applied to the query, key, value, output, gate, up, and down projections. At inference, C3R uses three deterministic option presentations, temperature $\tau = 0.1$, and the parameter-free argmax agreement rule. The checkpoint and inference configuration are fixed before development evaluation; no test-time threshold fitting or output repair is applied.

5 Results

Table 1 reports performance on the development and official test sets; C3R ranks first on the shared-task leaderboard. Relative to development, macro-F1 falls by 9.46%, whereas option accuracy falls by 4.25%, suggesting that factual recognition transfers more robustly than the cross-role decision.

Error analysis. Errors concentrate on responses that are only partially represented by the candidate set. Although cases without an exact response option constitute 55.5% of the test set, they account for 81.6% of label errors and 73.2% of option errors. The most difficult responses contain an additional qualifier or multiple claims that no single candidate captures. Three failure patterns emerge:

Split	Final	Macro-F1	Option
Dev.	0.9510	0.9619	0.9400
Test	0.8824	0.8673	0.8975

Table 1: C3R results with the ALLaM-7B backbone. "Final" is defined in Eq. 4.

the truth view can select an incorrect option even when cross-view disagreement yields the correct hallucination label; the match view can ignore an erroneous detail and produce accidental agreement; or both views can select the same semantically close distractor. Table 2 illustrates these failure modes.

ID	Outcome	Error case	Diagnosis
181	Label correct	A quotation by Zechariah is attributed to Moses.	T predicts the incorrect Moses option, while M selects a different option. Their disagreement correctly detects hallucination, but truth recovery fails.
268	Option correct	A valid definition of Sunnah divorce adds "one or two" pronouncements.	Both views select the gold one-pronouncement option, missing the added error.
240	Both wrong	Physical purification is confused with spiritual purification.	Both views select the same topically close but incorrect definition.

Table 2: Representative test errors. T and M denote the truth and match views.

6 Conclusion

We presented C3R, a structured recognition framework that jointly recovers the factual answer and detects hallucination through agreement between truth and response-matching roles. Counterfactual supervision, option permutations, and constrained likelihood scoring enable a QLoRA-adapted ALLaM-7B model to achieve a final score of 0.8824 and rank first on the official HalluScoring 2026 test leaderboard. The remaining errors mainly involve responses whose qualifiers or multiple claims are not fully captured by any single candidate, indicating that finer-grained response matching is a promising direction for future work.

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